Parakeet Café

SALES FORECAST · OCTOBER 2026

2027 Sales Forecast

Prophet, SARIMA and Holt-Winters, tuned and tested head-to-head on four years of Toast sales from our eleven cafés.



Toast POS net sales via MCP Andersons · 14,239 café-days · Sep 2022 – Oct 14, 2026
Prepared October 15, 2026

2027 sales plan: $32.5M, +12% on 2026

We forecast each café's sales with three time-series models, tuned their settings and features on 2025, confirmed the gains on January–September 2026, and recommend the blend of the three, the most accurate option in every round.

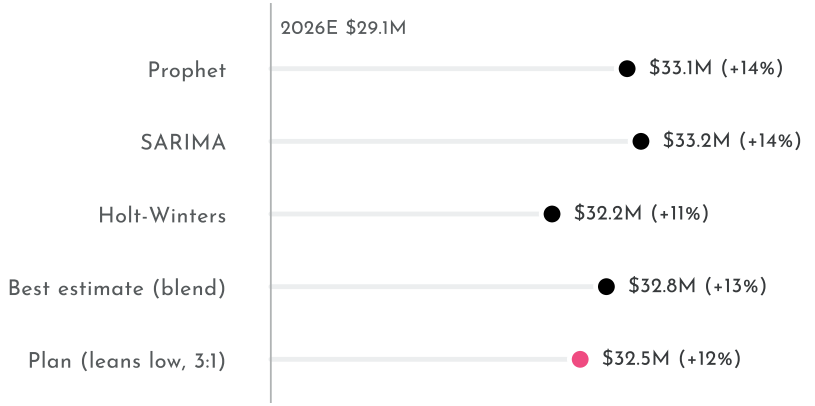
$32.5M

2027 sales plan · best estimate $32.8M shaded -0.9% because over-forecasting costs 3× more · existing 11 cafés

2027 best estimate	2026 estimate	2027 model range
$32.8M	**$29.1M**	**$32.2–33.2M**
+13% · blend of 3 tuned models	+15% vs 2025	+11% to +14%

2027 forecast by model

Net sales, 11 cafés · % change vs 2026 estimate



01 **The flock keeps growing.** 2026 is landing near $29.1M (+15%). Like-for-like sales are up +15% this year on +13% more checks: the growth is traffic, not price.

02 **The three models agree within a narrow band.** Prophet $33.1M, SARIMA $33.2M, Holt-Winters $32.2M: all expect another year of double-digit growth (+11% to +14%), now within $1.0M of each other (was $2.0M before tuning).

03 **Tuning sharpened the forecast.** After 294 settings were tried on 2025, the blend's error on the untouched Jan–Sep 2026 holdout fell from 8.8% to 8.6% per café-month and from 2.6% to 2.4% for the brand total.

04 **Plan on $32.5M, leaning low on purpose.** Over-forecasting is 3× costlier, so the plan shades the best estimate by -0.9%; in 2026 that would have cut the months where the plan beat actual sales from 33% to 21%. New cafés are not in the number.

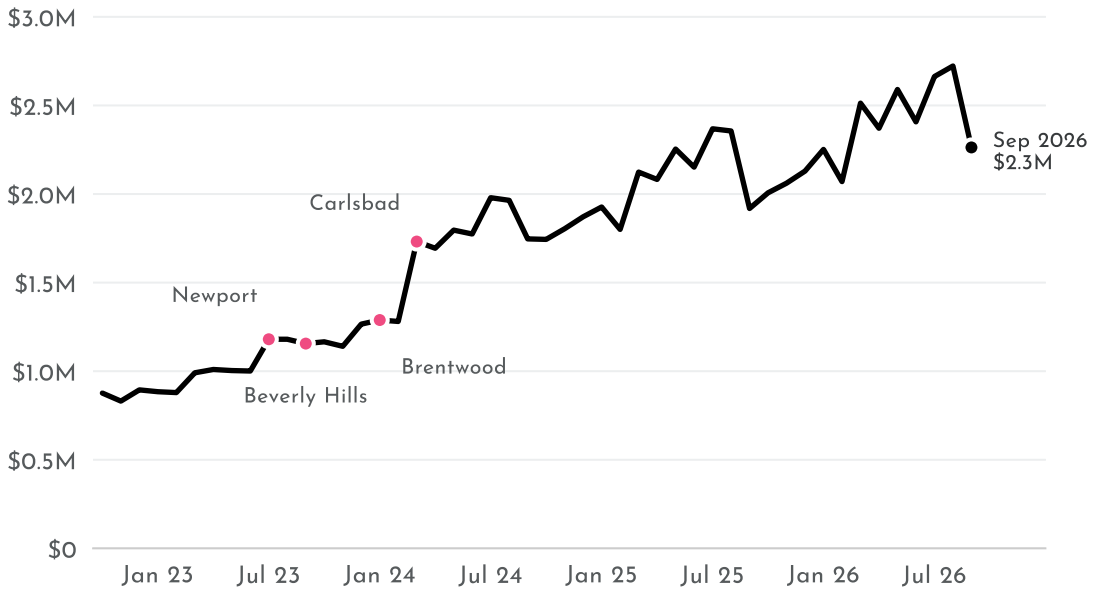
LIVE CHECK October 1–14, 2026, after the data the models were trained on: Toast recorded $1.08M; the best estimate was -1.4% off and the plan -2.2% (Prophet -0.7%, SARIMA -0.7%, Holt-Winters -2.7%).

Four years of growth: first new cafés, now the same cafés

Sales stepped up as Newport, Beverly Hills, Brentwood and Carlsbad opened in 2023–24. No café has opened since March 2024, so nearly all of the gains in 2025 and 2026 came from the same eleven cafés selling more.

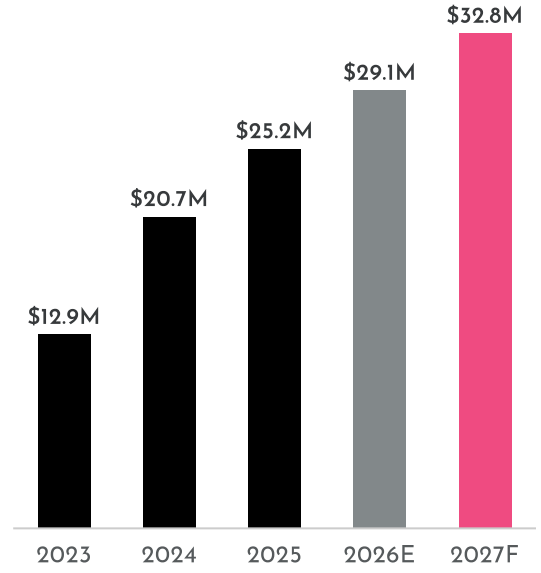
Monthly net sales, all cafés

Oct 2022 – Sep 2026 · pink dots mark café openings



Annual net sales

2026E = Jan–Sep actual + Q4 forecast · 2027F = blend



LIKE-FOR-LIKE GROWTH: SAME CAFÉS, SAME MONTHS

PERIOD	CAFÉS	NET SALES	CHECKS (TRAFFIC)	AVG. CHECK	AVG. CHECK ($)
Jan–Sep 2024 vs 2023	7	**+10.4%**	+0.1%	+10.4%	$23.31
Jan–Sep 2025 vs 2024	9	**+19.6%**	+11.3%	+7.5%	$25.81
Jan–Sep 2026 vs 2025	11	**+15.1%**	+13.2%	+1.7%	$26.64

The mix of growth has shifted. In 2024 it was all price (+10% average check, flat traffic). In 2026 it is almost all traffic (+13% checks, +2% average check). Busier cafés are a sturdier base for 2027 than price alone.

Like-for-like = cafés trading the full Jan–Sep period in both years · checks = Toast checks

Three ways to read the same history

Each model is fitted to every café separately and the eleven forecasts are added up, so a new opening is never mistaken for organic growth. The models differ mainly in how they extend the trend.

REQUESTED MODEL · META

Prophet

Splits daily sales into a trend, a weekly rhythm, a yearly season and holiday effects, then extends each one.

SEES

Daily sales per café, so weekends and holidays are learned directly (Mother's Day +59%, Memorial Day +39%, Christmas Eve −17%).

ASSUMES ABOUT GROWTH

The recent trend continues in a straight line, adding similar dollars each year.

WATCH-OUT

Ran about 2% hot on average in the round-1 back-tests.

ROUND-1 BACK-TEST

9.7% café-month error · 3.6% brand monthly error

2026 HOLDOUT, SETTINGS USED

9.7% café-month error · 2.4% brand error (tuned)

$33.1M 2027 · +14%

CLASSIC STATISTICS · BOX-JENKINS

SARIMA

Seasonal ARIMA: next year looks like last year, adjusted by the recent trend in the café's year-on-year changes.

SEES

Monthly sales per café, adjusted for how many Saturdays, Sundays and closed days each month has.

ASSUMES ABOUT GROWTH

Growth follows the café's recent year-on-year trend, re-estimated as each month lands, rather than its long-run average.

WATCH-OUT

The least precise café by café in round 1; the highest 2027 forecast.

ROUND-1 BACK-TEST

11.6% café-month error · 3.4% brand monthly error

2026 HOLDOUT, SETTINGS USED

8.5% café-month error · 2.7% brand error (tuned)

$33.2M 2027 · +14%

CLASSIC STATISTICS · EXPONENTIAL SMOOTHING

Holt-Winters

Tracks a level, a trend and a seasonal pattern, giving the latest months the most weight.

SEES

The same trading-day-adjusted monthly sales as SARIMA.

ASSUMES ABOUT GROWTH

A damped trend: growth continues but gradually flattens out.

WATCH-OUT

Under-forecast in four of five round-1 back-tests; the lowest 2027 forecast.

ROUND-1 BACK-TEST

9.4% café-month error · 3.3% brand monthly error

2026 HOLDOUT, SETTINGS USED

9.7% café-month error · 4.1% brand error (round-1 settings kept)

$32.2M 2027 · +11%

SAME DATA

Toast net sales for 11 cafés, Sep 2022 to Sep 2026; models fitted on data through Sep 30, 2026.

SAME CLEAN-UP

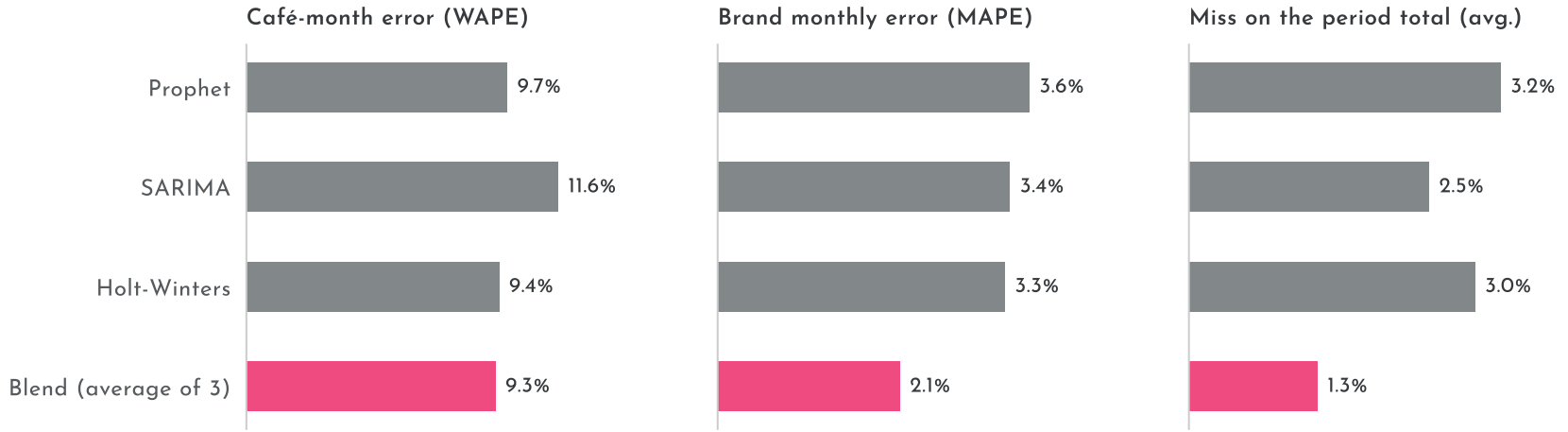
One-off disruptions are removed from training (Tropical Storm Hilary, Aug 2023; the LA fires at Brentwood, Jan 2025; closure days) but still count as actual sales in every test.

SAME SCORING

Every model faces the same five back-tests and is judged against real Toast sales.

Five dress rehearsals: which model would have called it?

We rewound the clock five times (Sep 2024, Dec 2024, Mar 2025, Jun 2025, Sep 2025). Each time, every model saw only the data known on that date and forecast months 4 to 15 ahead, the same gap that separates today from 2027. That gave 393 café-months to score against what Toast actually recorded.



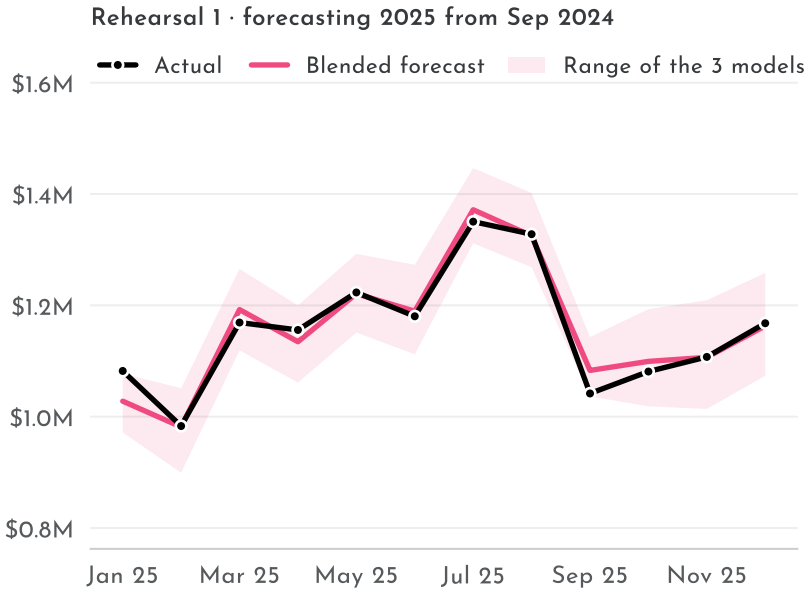
MODEL	CAFÉ-MONTH ERROR (WAPE)	BRAND MONTHLY ERROR (MAPE)	AVG. MISS ON PERIOD TOTAL	WORST MISS ON PERIOD TOTAL	AVERAGE BIAS
Prophet	9.7%	3.6%	3.2%	6.8%	+2.2% over
SARIMA	11.6%	3.4%	2.5%	6.0%	-1.8% under
Holt-Winters	9.4%	3.3%	3.0%	5.7%	-2.3% under
Blend (average of 3) BEST	**9.3%**	**2.1%**	**1.3%**	**2.5%**	**-0.6% under**

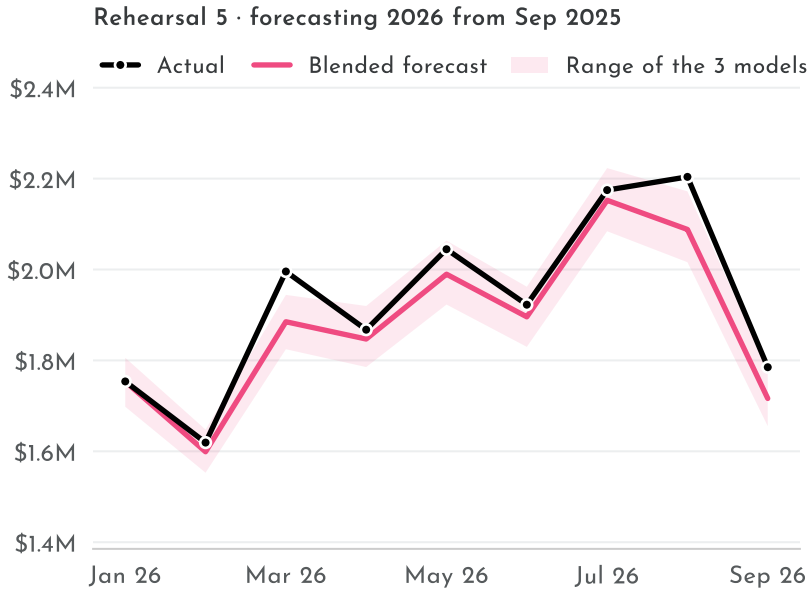
Lower is better. WAPE = total absolute error ÷ total sales, over all café-months. MAPE = average % error of the 11-café monthly total. Period total = the 12 months each rehearsal covered (9 for the last one).

Verdict. Holt-Winters is the most accurate single model month to month, a nose ahead of Prophet; the two are effectively tied café by café. SARIMA is the best of the three on period totals but the least precise café by café. Prophet tends to run a little hot and Holt-Winters a little cold, so their errors partly cancel. **The blend beat every single model on every measure**, and its worst miss on a period total was 2.5%.

The rehearsals up close

Rehearsals 1 and 5 bracket the test: the oldest forecast, made with two years of history, and the newest. The shaded band is the spread between the three models; the pink line is their average.





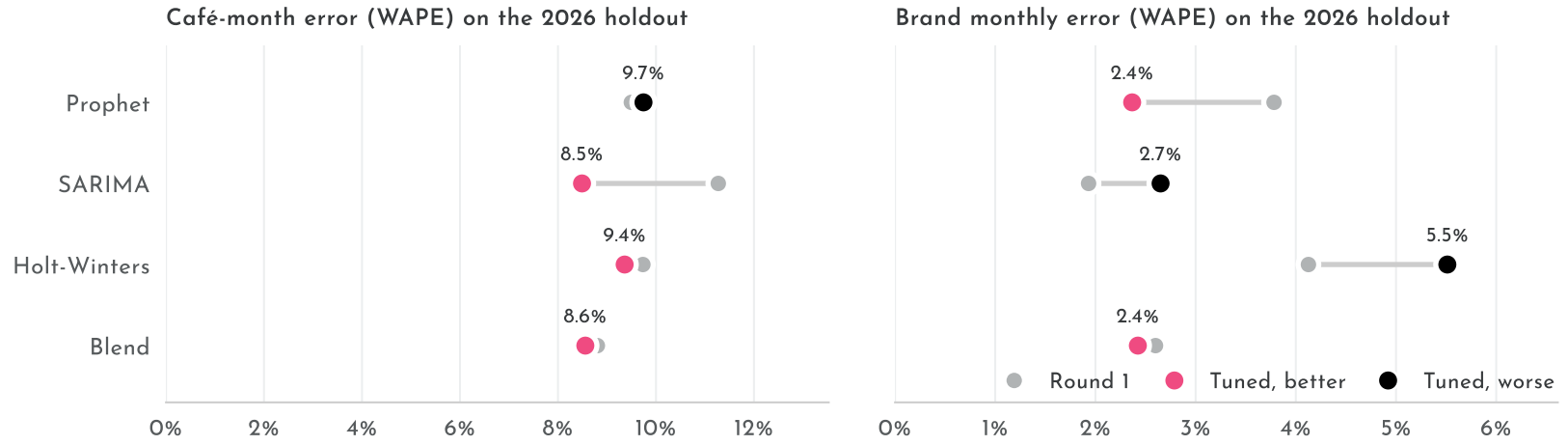
MISS ON THE PERIOD TOTAL, BY REHEARSAL

REHEARSAL (DATA THROUGH)	CAFÉS	MONTHS	PROPHET	SARIMA	HOLT-WINTERS	BLEND
1 · Sep 30, 2024	5	12	+6.8%	-6.0%	-0.2%	**+0.2%**
2 · Dec 31, 2024	7	12	+2.8%	-3.3%	-4.6%	**-1.7%**
3 · Mar 31, 2025	7	12	+1.8%	-1.5%	-2.5%	**-0.7%**
4 · Jun 30, 2025	7	12	+2.0%	+0.9%	+1.8%	**+1.6%**
5 · Sep 30, 2025	9	9	-2.7%	+0.9%	-5.7%	**-2.5%**

Live check, October 1–14, 2026. These two weeks came after the data the final models were trained on. Toast recorded $1.08M; the blend forecast was −1.4% off (Prophet −0.7%, SARIMA −0.7%, Holt-Winters −2.7%).

Tuned on 2025, proven on 2026

Each model searched for better settings and features using only its forecasts of 2025: 150 Prophet trials (Optuna) and every one of 48 SARIMA and 96 Holt-Winters combinations. The winners then faced a holdout the search never saw: forecasts made in June, September and December 2025 for January to September 2026 (198 café-months).



MODEL	CAFÉ-MONTH ERROR ROUND 1 → TUNED	BRAND MONTHLY ERROR ROUND 1 → TUNED	MISS ON PERIOD TOTAL ROUND 1 → TUNED	DECISION
Prophet	9.5% → **9.7%**	3.8% → **2.4%**	3.7% → **1.5%**	Adopt tuned settings
SARIMA	11.3% → **8.5%**	1.9% → **2.7%**	0.9% → **2.3%**	Adopt tuned settings
Holt-Winters	9.7% → **9.4%**	4.1% → **5.5%**	3.6% → **5.0%**	Keep round-1 settings
Blend	**8.8% → 8.6%**	**2.6% → 2.4%**	**2.3% → 2.0%**	**Inverse-error weights**

Holdout: Jan–Sep 2026 sales, never used to choose settings. A tuned model is adopted only if it lowers the holdout score (average of café-month and brand-month error).

Verdict. On the 2026 holdout score, tuning won for Prophet and SARIMA. Prophet improved its brand error 3.8% → 2.4% while giving back a little at the other level (café-month 9.5% → 9.7%); SARIMA improved its café-month error 11.3% → 8.5% while giving back a little at the other level (brand 1.9% → 2.7%). Holt-Winters keeps round-1 settings: the tuned version looked better on 2025 but did worse on 2026. **The blend** moves from 8.8% to 8.6% café-month error and from 2.6% to 2.4% at brand level; its average miss on the period total is 2.0% (was 2.3%).

Which features earned their place

Starting from each model's best settings, every feature was switched on and off on its own and scored on 2025 and on the 2026 holdout. What helped: multiplicative rather than additive seasonality (Prophet), weekday-weighted trading days (SARIMA) and multiplicative rather than additive seasonality (Holt-Winters). What hurt: holiday periods (Prophet) and log transform (SARIMA).

MODEL	FEATURE	EFFECT ON 2025 SCORE	EFFECT ON 2026 HOLDOUT	VERDICT
Prophet	Holiday periods: Christmas week, spring break, Thanksgiving weekend	+4%	+4%	Hurts
Prophet	Separate weekly pattern for summer	+6%	-1%	Mixed
Prophet	Cap catering spikes at 3× the 29-day median	-0%	+0%	No difference
Prophet	Multiplicative rather than additive seasonality · used	-17%	-4%	**Helps**
SARIMA	Weekday-weighted trading days (vs plain day count) · used	-2%	-3%	**Helps**
SARIMA	Log transform	+15%	+13%	Hurts
Holt-Winters	Weekday-weighted trading days (vs plain day count) · used	-15%	+0%	Mixed
Holt-Winters	Log transform	+3%	-4%	Mixed
Holt-Winters	Multiplicative rather than additive seasonality · used	-24%	-14%	**Helps**

Each feature switched on vs off with every other setting at the model's best; effect = change in score (average of café-month and brand-month error) when it is on, so negative is better. "Helps" or "Hurts" only when 2025 and 2026 agree; "No difference" when both move less than 1%. Weather could not be tested: this environment's network policy blocks the weather archive.

HOW MUCH GROWTH TO EXTRAPOLATE?

TREND SETTING	BEST SCORE
SARIMA · Local trend (re-estimated)	5.56%
SARIMA · Constant growth (drift)	6.60%
SARIMA · No growth term	6.94%
Holt-Winters · Damped trend, φ = 0.95	4.56%
Holt-Winters · Damped trend, φ = 0.98	4.63%
Holt-Winters · Damped trend, φ = 0.9	5.19%
Holt-Winters · Damped trend, φ estimated	5.22%
Holt-Winters · Straight-line trend	5.23%
Holt-Winters · No trend	8.76%

One model per café, or one model for all?

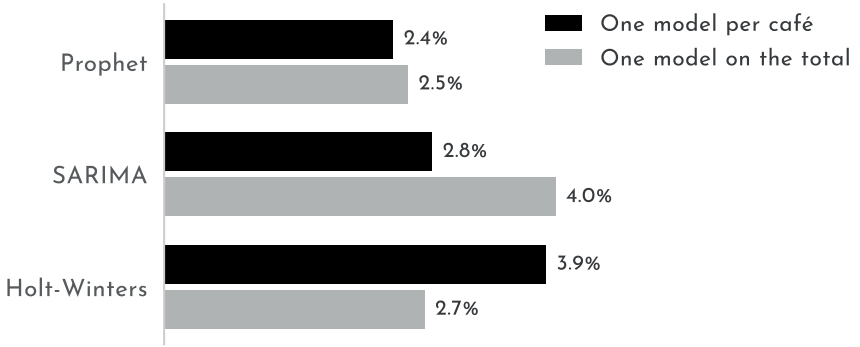
Round 1 fitted every model once per café and added the results up. Here each model is also fitted once to the cafés' combined sales, and both are scored on the same 2026 holdout.

MODEL (TUNED SETTINGS)	BRAND MONTHLY ERROR PER CAFÉ	BRAND MONTHLY ERROR ONE MODEL	MISS ON PERIOD TOTAL PER CAFÉ	MISS ON PERIOD TOTAL ONE MODEL	2027 FORECAST PER CAFÉ	2027 FORECAST ONE MODEL
Prophet	2.4%	2.5%	1.5%	2.0%	$33.07M	$32.68M
SARIMA	2.8%	4.0%	2.3%	3.4%	$33.23M	$33.48M
Holt-Winters	3.9%	2.7%	3.6%	2.5%	$32.22M	$32.08M

Holdout Jan–Sep 2026. "One model" is fitted to the combined sales of the cafés in each test; for 2027 it only has history from April 2024, when the eleventh café opened.

Brand monthly error on the 2026 holdout

Lower is better · tuned settings



One model per café wins for 2 of 3 models on brand monthly error. For 2027 the single models land within −1.2% to +0.8% of the per-café totals, so the headline barely depends on the choice. We keep one model per café: it is at least as accurate, it gives every café its own forecast, and it is not thrown off when a café opens (the combined series only starts in April 2024, when the eleventh café opened).

Brand monthly error = MAPE of the combined monthly sales of the cafés in each test

Leaning low on purpose

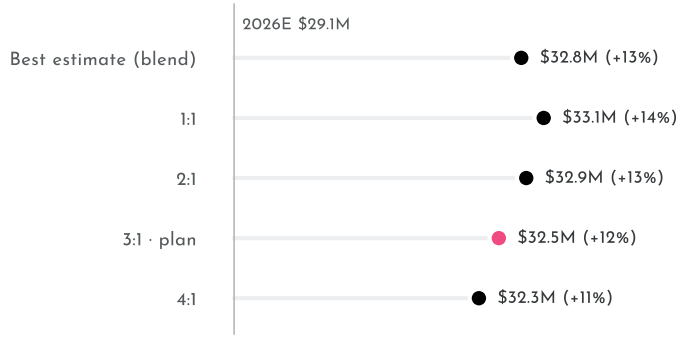
Over-forecasting sales costs more than under-forecasting: labor, purchasing and targets get set too high. So the plan sits below the best estimate. The models were not weakened to get there; they are still tuned only for accuracy. Instead we measured how the blend has actually missed, and set the plan where the expected cost is lowest when an over-forecast dollar costs 3 under-forecast dollars.

IF OVER-FORECASTING COSTS	SHADE THE BEST ESTIMATE	2027 PLAN	2026 MONTHS ABOVE ACTUAL BEST ESTIMATE → PLAN	2026 COST BEST ESTIMATE → PLAN
1× under-forecasting	+0.9%	$33.13M	33% → 42%	2.4% → 2.2%
2× under-forecasting	+0.2%	$32.90M	33% → 33%	3.1% → 3.1%
3× under-forecasting PLAN	**-0.9%**	**$32.54M**	**33% → 21%**	**3.8% → 3.9%**
4× under-forecasting	-1.7%	$32.28M	33% → 21%	4.5% → 4.3%

Shade = the multiplier that minimises the asymmetric cost of monthly brand misses on forecasts made 4–15 months ahead (2025 and 2026 targets). The 2026 columns test the multiplier fitted on 2025 alone against Jan–Sep 2026 (24 brand-months). Cost = over-forecast dollars × ratio + under-forecast dollars, as a share of sales.

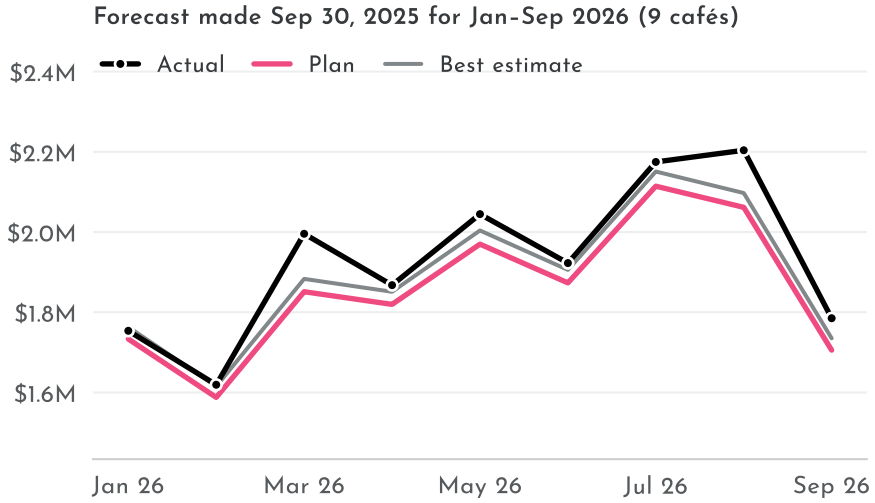
2027 plan by cost of over-forecasting

Net sales, 11 cafés · % vs 2026 estimate



How the plan would have tracked 2026

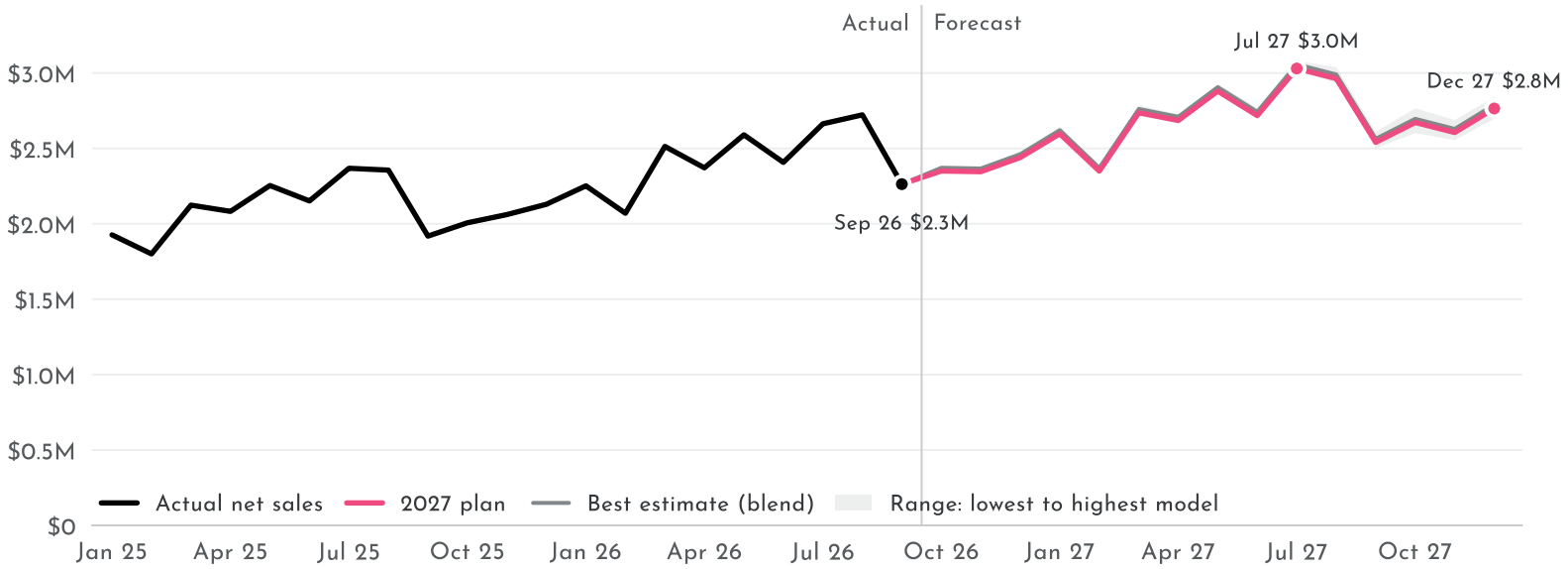
Plan = best estimate × the 3:1 shade fitted on 2025 only



What the evidence says. The tuned blend already leans slightly low: on the 2026 holdout it came in 1.0% under actual sales and above them in 33% of months. So the 3:1 shade is modest, −0.9% overall: −0.2% for months 4–9 ahead and −1.8% for months 10–15 ahead, where misses grow. In 2026 it cut the months where the plan beat actual sales from 33% to 21%; because 2026 beat the forecast, it cost slightly more than it saved that year (3.8% → 3.9%). It is insurance for the year that disappoints. Single cafés miss by far more than the total, so a café target each café should beat three months in four needs about −6%; such café targets add up to less than the brand plan by design.

2027, month by month

The seasonal shape holds: a March lift, a July–August summer peak, then the September dip. Every month of 2027 is forecast 9%–15% above the same month of 2026.



$ THOUSANDS	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	2027
Prophet	2,645	2,385	2,763	2,721	2,922	2,769	3,080	3,001	2,581	2,723	2,647	2,833	$33.07M
SARIMA	2,626	2,384	2,786	2,728	2,926	2,762	3,071	3,038	2,611	2,766	2,688	2,841	$33.23M
Holt-Winters	2,590	2,344	2,736	2,680	2,871	2,696	3,020	2,933	2,501	2,602	2,555	2,697	$32.22M
Best estimate	2,620	2,371	2,761	2,710	2,906	2,742	3,057	2,990	2,563	2,696	2,629	2,790	$32.84M
Plan	**2,597**	**2,349**	**2,736**	**2,685**	**2,880**	**2,717**	**3,030**	**2,963**	**2,540**	**2,672**	**2,606**	**2,765**	**$32.54M**
2026 (Q4 estimate)	2,252	2,070	2,513	2,371	2,590	2,407	2,663	2,723	2,263	2,371	2,367	2,462	$29.05M
Plan vs 2026	+15%	+13%	+9%	+13%	+11%	+13%	+14%	+9%	+12%	+13%	+10%	+12%	+12%

2027, café by café

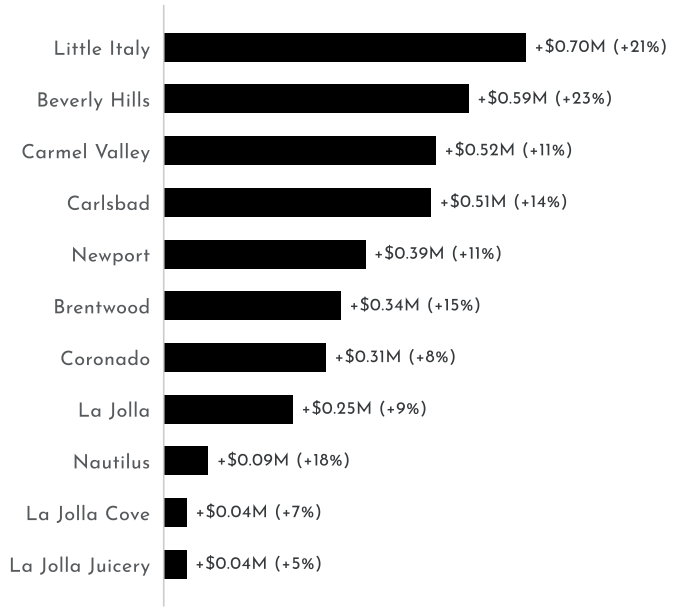
Little Italy, Beverly Hills and Carmel Valley deliver 48% of the $3.8M of growth. La Jolla Juicery (+5%) and La Jolla Cove (+7%) grow the least.

CAFÉ	REGION	2025 ACTUAL	2026 ESTIMATE	2027 BEST ESTIMATE	VS 2026	2027 PLAN	MODEL RANGE (LOW–HIGH)
Carmel Valley	San Diego	4.24	4.67	5.20	+11%	**5.15**	5.13–5.31
Coronado	San Diego	3.53	4.03	4.35	+8%	**4.31**	3.98–4.67
Carlsbad	San Diego	3.34	3.76	4.27	+14%	**4.23**	4.15–4.35
Little Italy	San Diego	2.57	3.27	3.96	+21%	**3.93**	3.90–4.09
Newport	Orange County	3.18	3.50	3.89	+11%	**3.85**	3.85–3.92
La Jolla	San Diego	2.74	2.91	3.16	+9%	**3.13**	3.14–3.19
Beverly Hills	Los Angeles	1.91	2.56	3.15	+23%	**3.12**	3.06–3.20
Brentwood	Los Angeles	1.87	2.32	2.67	+15%	**2.64**	2.53–2.76
La Jolla Juicery	San Diego	0.79	0.90	0.94	+5%	**0.93**	0.89–0.97
La Jolla Cove	San Diego	0.65	0.64	0.68	+7%	**0.68**	0.65–0.72
Nautilus	San Diego	0.37	0.49	0.57	+18%	**0.57**	0.55–0.60
All cafés		**25.18**	**29.05**	**32.84**	**+13%**	**32.54**	**32.22–33.23**

$ millions, net sales. Model range = lowest and highest of Prophet, SARIMA and Holt-Winters for that café (café ranges do not add up to the total range).

Where the extra $3.8M comes from

2027 forecast minus 2026 estimate, by café



What could ruffle the forecast

The models know only the past. They see no new cafés, price changes or local events, and they assume the habits of 2024–26 continue. These four things would move the 2027 number.

NEW CAFÉS ARE UPSIDE

+$2.0M

average first-year sales of a new café

Manhattan Beach is set up in Toast and Studio City and Enclave in R365, but none has sales yet, so none is in the forecast. Recent openings did Newport $2.4M, Beverly Hills $1.4M, Brentwood $1.5M, Carlsbad $2.9M in their first 12 months. Add each one, prorated by the months it trades in 2027.

MOMENTUM IS THE BIG BET

$31.4M

2027 if like-for-like growth cools to +8%

All three models extend like-for-like growth of +10%, +20% and +15% (2024–26). If 2027 growth slows to +8%, sales land near $31.4M, below the lowest model. Track like-for-like monthly from January.

GROWTH IS CONCENTRATED

48%

of 2027 growth comes from three cafés

Little Italy (+21%, after a step up in March 2026) and Beverly Hills (+23%) lead. If both grew +10% instead, 2027 would land near $32.1M (-2.1% vs the blend).

PRICE IS AN OPEN LEVER

+1.7%

average check, Jan–Sep 2026 vs 2025

2026 growth is traffic. The average check was nearly flat at $26.64. The models only extend sales trends and know nothing about menu prices, so a 2027 price increase is a lever on top of this forecast, provided traffic holds.

2027 under different assumptions

Net sales, existing 11 cafés unless stated · % vs 2026 estimate



HOW TO USE IT

Budget at the plan, **$32.5M**; the best estimate is $32.8M. For café targets each café should beat three months in four, shade café forecasts by about -6% (single cafés miss by more). Add new cafés on top as their opening dates firm up.

WATCH FIRST

Q1 2027 like-for-like against the forecast's +12%. A gap of more than 3 points is the signal to revisit.

KEEP IT FRESH

Re-run monthly as Toast data lands; the full pipeline (extract, back-test, forecast, this report) runs in minutes from the repo.

* Illustrative: half of the average first-year sales of our last four openings ($1.0M); opening dates are not yet set

Method and data

DATA

Toast POS tables in the Grupo Anderson's BI warehouse, read through MCP Andersons. Net sales follow the official *silver.v_toast_sales_daily* definition: check amounts on non-voided, non-deleted checks and orders, minus Toast gift-card sales. 14,239 café-days ($84.8M) from Sep 7, 2022 to Oct 14, 2026, validated month by month against checksums computed in SQL.

MODELLING WINDOW

Toast history starts with the Sep 2022 POS migration, so modelling starts Oct 2022. Cafés opened since then skip their 2–4 week opening ramp. Nautilus (a weekday corporate-campus café) trades Monday–Friday and closes on five holidays; the forecasts respect that calendar.

ONE-OFF DISRUPTIONS

- All cafés: Aug 20, 2023 (Tropical Storm Hilary closures)
- Brentwood: Jan 7–25, 2025 (Los Angeles wildfires)
- La Jolla: Jul 8, 2025 (unplanned closure)
- La Jolla Juicery: Jul 8, 2025 (unplanned closure)
- Little Italy: Jun 11–18, 2025 (closures)
- Little Italy: Sep 29–30, 2025 (closures)
- Any other day below 25% of the café's 29-day median (partial closures)

Excluded from Prophet's training; imputed (same weekday, nearby weeks) for the monthly models; kept in the actuals used for scoring.

PROPHET

Daily net sales per café: linear trend, weekly and yearly seasonality, US federal holidays plus Valentine's Day, Easter, Mother's and Father's Day, Black Friday, Christmas Eve and New Year's Eve. Monthly totals = sum of daily forecasts on trading days.

SARIMA

Monthly sales rate per café (sales ÷ trading days), seasonal ARIMA with the order chosen per café by an information criterion; orders used for 2027 are listed in the next appendix.

HOLT-WINTERS

Same monthly rate; exponential smoothing with a trend and a 12-month seasonal pattern, parameters estimated by least squares.

BLEND

Combines the three café by café and month by month (inverse-error weights). Settings for 2027 are on the next page.

TUNING DESIGN (ROUND 2)

Tuning set. Forecasts made at the end of Sep 2024, Dec 2024, Mar 2025 and Jun 2025, scored on months 4–15 ahead that fall in 2025 (186 café-months).

Holdout. Forecasts made at the end of Jun, Sep and Dec 2025, scored only on Jan–Sep 2026 (198 café-months), never used to choose anything.

Score. Average of café-month and brand-month error (WAPE), so a setting cannot win by sharpening cafés while drifting on the total.

Search. Prophet: 150 Optuna (TPE) trials over nine settings and feature switches, starting from round 1. SARIMA: all 48 combinations. Holt-Winters: all 96.

Rules. A model's tuned settings are adopted only if they lower its holdout score. The blend rule (average, median or error-weighted) is picked on the tuning set and kept only if it matches the average on the holdout.

2027 forecast by café and model

CAFÉ	REGION	2025 ACTUAL	2026 ESTIMATE	PROPHET	SARIMA	HOLT-WINTERS	BLEND	PLAN	SARIMA ORDER CHOSEN
Carmel Valley	San Diego	$4.24M	$4.67M	$5.15M	$5.13M	$5.31M	**$5.20M**	$5.15M	SARIMA(0, 1, 1)(0, 1, 0)12
Coronado	San Diego	$3.53M	$4.03M	$4.67M	$4.38M	$3.98M	**$4.35M**	$4.31M	SARIMA(1, 1, 0)(0, 1, 1)12
Carlsbad	San Diego	$3.34M	$3.76M	$4.15M	$4.35M	$4.33M	**$4.27M**	$4.23M	SARIMA(0, 1, 0)(0, 1, 1)12
Little Italy	San Diego	$2.57M	$3.27M	$3.91M	$4.09M	$3.90M	**$3.96M**	$3.93M	SARIMA(0, 1, 0)(0, 1, 0)12
Newport	Orange County	$3.18M	$3.50M	$3.92M	$3.90M	$3.85M	**$3.89M**	$3.85M	SARIMA(0, 1, 0)(0, 1, 0)12
La Jolla	San Diego	$2.74M	$2.91M	$3.14M	$3.19M	$3.14M	**$3.16M**	$3.13M	SARIMA(1, 1, 0)(0, 1, 1)12
Beverly Hills	Los Angeles	$1.91M	$2.56M	$3.20M	$3.19M	$3.06M	**$3.15M**	$3.12M	SARIMA(1, 1, 1)(0, 1, 0)12
Brentwood	Los Angeles	$1.87M	$2.32M	$2.76M	$2.70M	$2.53M	**$2.67M**	$2.64M	SARIMA(0, 1, 1)(0, 1, 0)12
La Jolla Juicery	San Diego	$0.79M	$0.90M	$0.96M	$0.97M	$0.89M	**$0.94M**	$0.93M	SARIMA(0, 1, 1)(0, 1, 1)12
La Jolla Cove	San Diego	$0.65M	$0.64M	$0.65M	$0.72M	$0.68M	**$0.68M**	$0.68M	SARIMA(0, 1, 0)(0, 1, 1)12
Nautilus	San Diego	$0.37M	$0.49M	$0.55M	$0.60M	$0.57M	**$0.57M**	$0.57M	SARIMA(1, 1, 0)(0, 1, 1)12
All cafés		**$25.18M**	**$29.05M**	**$33.07M**	**$33.23M**	**$32.22M**	**$32.84M**	**$32.54M**	

FILES IN THE REPO

- *data/raw/parakeet_toast_daily.csv*: daily net sales and checks by café
- *outputs/forecast_by_store_month.csv*: Oct 2026 – Dec 2027 by café, month and model
- *outputs/forecast_brand_month.csv*: brand totals with simulation ranges
- *outputs/backtest_forecasts.csv*, *backtest_metrics.csv*: every rehearsal forecast and score

READING THE RANGES

The low–high model range is the practical planning band. Each model's own 80% interval for the 2027 total (in *forecast_annual.csv*) ranges from ±0.3% for Prophet to ±7.0% for Holt-Winters. They disagree too much, and Prophet's is far tighter than its back-test misses, so we don't headline them.

REPRODUCE

python src/backtest.py → *python src/tune.py* → *python src/forecast_2027.py* → *python report/build_report.py*. Data refresh: *data/sql/extract_parakeet_daily.sql* via MCP Andersons.

Settings: round 1 vs tuned

MODEL	SETTING	BEST FOUND ON 2025	ROUND 1	USED FOR 2027
Prophet	**Trend flexibility**	**0.00846**	**0.05**	**0.00846**
	Trend can bend until	**74% of history**	**80% of history**	**74% of history**
	Seasonality strength	**0.0177**	**10**	**0.0177**
	Holiday strength	**2.81**	**10**	**2.81**
	Seasonality	multiplicative	multiplicative	multiplicative
	Yearly detail (Fourier terms)	**20**	**10**	**20**
	Holiday periods	off	off	off
	Summer weekly pattern	off	off	off
	Cap catering spikes	off	off	off
SARIMA	**Growth term**	**Local trend (re-estimated)**	**Constant growth (drift)**	**Local trend (re-estimated)**
	Orders searched	**12 (extended)**	**6**	**12 (extended)**
	Order chosen by	AICc	AICc	AICc
	Transform	**none**	**log**	**none**
	Trading days	weekday	weekday	weekday
Holt-Winters	**Trend**	**Damped trend, φ = 0.95**	**Damped trend, φ estimated**	**Damped trend, φ estimated**
	Seasonality	multiplicative	multiplicative	multiplicative
	Transform	none	none	none
	Starting values	estimated	estimated	estimated
	Trading days	weekday	weekday	weekday

Bold rows changed. "Best found" is the top score on the 2025 tuning set; it is used for 2027 only where it also won on the 2026 holdout (Prophet and SARIMA). Trend flexibility, seasonality and holiday strength are Prophet's prior scales: higher lets that component bend more. Blend weights: Prophet 35%, SARIMA 31%, Holt-Winters 33%.